

Improve performance of existing online trackers for small object detection [UAV videos]

Team: The Convolutioners

Aagam Sheth
AU2544006
M.Tech (CSE)

Mahima Parekh
AU2544003
M.Tech (CSE)

Khevan Thanki
AU23L20002
BS (Hons) CS

Riddhi Bhargava
AU22220028
BS (Hons) CS

Abstract—Small object detection (SOD) and multi-object tracking (MOT) in unmanned aerial vehicle (UAV) videos are important research topics in computer vision. Detecting small objects in different environments has been challenging because of small object sizes, low frame resolutions, and changing scene conditions. These issues often cause missed detections and more false positives in current object detectors. To address this, the project uses a deep learning framework for reliable SOD and MOT in UAV videos. It incorporates SAHI, a tiling strategy, to improve the performance of the You Only Look Once (YOLO) detector on the VisDrone2019-MOT dataset. A custom tracking module is developed to improve re-identification by using a Siamese Network Module and Kalman filter for motion prediction. The Siamese Re-ID module is systematically optimized using a 12-configuration grid sweep, 12-trial Bayesian optimization with Optuna, and an alternative Focal-BCE loss formulation with asymmetric negative weighting, which reduces false positives by 21.5%. This project aims to provide a modular framework for evaluating tracking performance in different UAV scenarios and to test the proposed system against established benchmarks.

Index Terms—Small Object Detection, Multi-Object Tracking, UAV, VisDrone, Siamese Network, Contrastive Loss, Focal-BCE, Hyperparameter Optimisation, Optuna, Kalman Filter.

I. LITERATURE REVIEW

The detection of small objects in complex visual scenes remains a significant challenge. This issue is recognized as one of the most persistent challenges in the field of computer vision. The problem is particularly pronounced in aerial and unmanned aerial vehicle (UAV) imagery, where even modest changes in the altitude of the vehicle can result in substantial variations in the appearance of objects within an image [1]. The receptive fields of standard convolutional neural network architectures are often inadequate for detecting small objects, as their receptive fields typically cover only a few pixels. Diverse and cluttered backgrounds further exacerbate this issue, making it challenging to distinguish genuine foreground objects from background textures, particularly when larger, dominant objects are present within the same scene [2]. A foundational architectural response to the challenge of scale variance was the introduction of Feature Pyramid Networks (FPN) by [3]. Prior to FPN, multi-scale detection either processed the same image at multiple resolutions independently, which was computationally heavy, or relied on a single feature map, therefore degrading performance on smaller objects. FPN

addressed this by exploiting the hierarchical structure of deep convolutional networks where a top-down pathway merges spatially coarser but semantically richer features from higher layers with finer-resolution maps from lower layers via lateral connections and element-wise addition. The result is a set of feature maps at multiple scales, all enriched with high-level semantic information, enabling predictions to be made at each scale independently.

The YOLO family of single-stage detectors incorporated FPN-style neck architectures progressively through YOLOv3 and later versions. More recently, YOLOv11 achieves 44.1% mAP@0.5 on the VisDrone2019 dataset, positioning the YOLO family as competitive detectors even against Transformer-based detectors [1]. Despite these advances, YOLO architectures carry structural limitations for small object scenarios. Therefore, one of the most practical and architecture-agnostic approaches to improving small object detection is image tiling. Rather than feeding a downsampled full image to the detector, tiling subdivides the image into overlapping patches at the network's native input resolution, runs inference on each patch independently, and merges predictions back into the original image space via Non-Maximum Suppression (NMS). The Slicing Aided Hyper Inference (SAHI) framework formalises this procedure [4].

Extending detection into the temporal domain, multi-object tracking (MOT) requires the continuous identification and re-identification of multiple targets across video frames. Occlusions, similar appearances in crowded scenes, and non-linear motion patterns are the primary failure modes of such systems [5]. Re-identification (ReID) plays a key role in addressing these failures. Siamese network architectures have been a prominent approach to appearance-based matching [6]. A Siamese network is made up of two identical subnetworks with shared weights that process two inputs at the same time. In the context of Re-ID, this setup makes it possible to match a reappearing candidate detection to a previous identity by comparing appearance embeddings.

Contrastive loss [7] is the canonical training objective for Siamese metric learning, minimising same-pair distance while pushing different-pair distance beyond a margin. Focal loss [8], originally proposed for object detection, re-weights hard examples via a modulating factor $(1 - p)^\gamma$; adapting this

to binary pair classification results in a Focal-BCE formulation. This formulation enables precision-oriented training. For hyperparameter search, Bayesian optimisation using Tree-structured Parzen Estimators (TPE), as implemented in Optuna [9], provides an efficient alternative to exhaustive grid search. The primary contribution of this study is a multi-scale detection pipeline that integrates the SAHI tiling strategy with a fine-tuned YOLO11 detector, a systematically optimised Siamese Re-ID network, and a BoT-SORT-based tracker with Kalman filtering. All components are evaluated on the VisDrone2019-MOT benchmark.

The remainder of this paper is structured as follows. Section II describes the dataset and proposed methodology. Section III presents results. Section IV discusses findings and implications. Section V concludes the paper.

II. DATASET AND METHODOLOGY

The proposed methodology consists of five phases: data preparation, baseline evaluation, tiling strategy implementation on YOLO, Siamese network design and optimisation, and custom tracker development.

A. Phase 1: Data Preparation

The VisDrone2019-MOT dataset is used to train and test the designed architecture. The dataset contains ten object categories including pedestrian, bicycle, car, van, truck, and motor. VisDrone annotations are transformed into YOLO format, where each object's class ID and bounding box are normalised to image dimensions as centre coordinates (c_x, c_y, w, h) . Annotations marked as ignored regions and boxes with invalid dimensions are filtered out. For the Siamese network, object crops are extracted from the training sequences by localising each annotated bounding box and saving the corresponding image patch.

B. Phase 2: Baseline Evaluation

YOLOv8n was evaluated on the VisDrone dataset to establish baseline performance metrics, reflecting the performance of a standard detector on full-resolution UAV frames without any scale-aware preprocessing. Additionally, YOLO11 was fine-tuned on the VisDrone-MOT dataset to assess performance gains compared to the non-fine-tuned model.

C. Phase 3: Fine-tuned YOLO11 + SAHI Tiling Strategy

A key limitation of the baseline YOLO model is that full VisDrone frames are downsampled during training and inference, causing small objects to become effectively undetectable after resizing. To address this, a tiling-based preprocessing strategy was applied using the SAHI framework. Each input frame is divided into overlapping 640×640 crops, and the fine-tuned YOLO11 detector is applied independently to each tile. Multiple detections produced for the same object across adjacent tiles are resolved by applying Non-Maximum Suppression (NMS) to merge redundant bounding boxes.

D. Phase 4: Siamese Network for Re-Identification

1) *Architecture and Pair Construction*: The pipeline begins with extracting ground-truth object crops from VisDrone annotations. Pairwise training samples are constructed generating positive pairs (same identity across frames) and negative pairs (different identities, including visually similar categories). The training set contains 6000 pairs per epoch and the validation set contains 3000 pairs.

The Siamese network consists of two weight-sharing branches, each comprising a ResNet18 backbone [10] (ImageNet-pretrained) followed by a fully connected projection head mapping 512-dimensional pool features to an embedding of dimension d . Similarity is computed as the Euclidean distance between embeddings:

$$D(x_i, x_j) = \|\hat{f}(x_i) - \hat{f}(x_j)\|_2, \quad (1)$$

where $\hat{f}(x) = f(x)/\|f(x)\|_2$ when L2 normalisation is enabled. A pair is classified as a match if $D < \tau$.

2) *Loss Functions*: The primary training objective is contrastive loss [7]:

$$\mathcal{L}_{\text{con}} = y D^2 + (1 - y) \max(m - D, 0)^2, \quad (2)$$

where $y=1$ for positive pairs, $y=0$ for negative pairs, and m is the margin. As an alternative, a Focal Binary Cross-Entropy (Focal-BCE) formulation is proposed. A lightweight head maps the concatenated embeddings to a scalar probability p , trained with:

$$\mathcal{L}_{\text{FBCE}} = -[w_y y(1 - p)^\gamma \log p + (1 - y) p^\gamma \log(1 - p)], \quad (3)$$

with $\gamma=2.0$. Two variants are evaluated: a **Baseline** with $w_{\text{neg}}=1.0$ (symmetric), and an **Enhanced FP-Reduced** variant with $w_{\text{neg}}=2.0$ that explicitly penalises false positives. The precision-constrained selection policy (maximise recall subject to precision ≥ 0.80) is used for the enhanced variant, since spurious Re-ID matches propagate incorrect track identities across frames and compound MOTA degradation.

3) *Hyperparameter Search: Grid Sweep*: A structured 12-configuration grid sweep is conducted across: backbone $\in \{\text{ResNet18, ResNet34, MobileNetV3-Small}\}$; embedding dimension $d \in \{64, 128, 256\}$; learning rate $\in \{10^{-4}, 3 \times 10^{-4}\}$; contrastive margin $m \in \{0.8, 1.0, 1.2\}$; dropout $\in \{0.0, 0.2\}$; backbone freezing; and L2 normalisation. Each configuration is trained for 3 epochs on 6000 pairs (batch size 32) and evaluated across six thresholds $\tau \in \{0.5, 0.8, 1.0, 1.2, 1.5, 2.0\}$.

4) *Hyperparameter Search: Optuna Bayesian Optimisation*: A 12-trial Optuna study using the TPE sampler is run in the Focal-BCE regime. The search space covers: $d \in \{64, 128\}$; dropout $\in [0.0, 0.3]$; $lr \in [10^{-4}, 5 \times 10^{-4}]$; negative weight $\in [1.0, 3.0]$; L2 normalisation; and backbone freezing. The objective is validation F1 after 3 epochs.

E. Phase 5: BoT-SORT Tracker Design

The tracking pipeline combines motion prediction, spatial matching, and appearance-based Re-ID into a two-stage association strategy. At each frame, existing tracks are predicted

forward using a constant-velocity Kalman filter, and incoming detections are split into high-confidence and low-confidence groups. In the first stage, high-confidence detections are matched to all active tracks using a blended cost of IoU distance and cosine appearance distance, resolved via the Hungarian algorithm. In the second stage, remaining unmatched tracks are associated with low-confidence detections using IoU cost alone. Stale tracks are pruned while unmatched detections satisfying birth thresholds are initialised as new tracks. To compensate for ego-motion introduced by the UAV platform, Camera Motion Compensation (CMC) is applied between consecutive frames using sparse optical flow. The estimated affine transformation is used to warp existing track bounding box predictions into the current frame’s coordinate system before the IoU and appearance matching stages. This is especially important in UAV footage where even slow drone drift causes all predicted track positions to shift simultaneously, inflating IoU distances and producing spurious mismatches without correction. To systematically improve tracker performance, a structured four-phase hyperparameter sweep was implemented, with all trials logged to `hparam_trials.csv` for reproducibility. Candidates are ranked by a composite score $= 0.45 \cdot \text{IDF1} + 0.35 \cdot \text{HOTA} + 0.20 \cdot \text{MOTA}$, prioritising identity consistency. Phase 1 swept association gates (`appearance_weight`, `match_cost_thresh`, `min_iou_gate`, `assoc_iou_thresh`) over 10 trials, yielding `appearance_weight=0.65`, `match_cost_thresh=0.72` as the optimal pair. Phase 2 swept track lifecycle parameters (`min_hits`, `max_age`, `new_track_thresh`) over 24 trials; `min_hits=1` proved decisive for UAV footage where immediate track admission is critical for fast-appearing small objects. Phases 3 (Kalman dynamics) and 4 (local reassociation) remain pending and represent the primary avenue for further tracker improvement. The Siamese network is central to the appearance matching component: each detection patch is resized to 128×128 and passed through one branch to produce a 128-dimensional L2-normalised embedding. Track embeddings are maintained via exponential moving average. YOLOv8 serves as the detection front-end, fully decoupled from the tracker.

III. RESULTS

A. Baseline Evaluation

TABLE I: YOLO11 + SAHI performance with existing BoT-SORT

Metric	Value
MOTA	17.28%
MOTP	26.01%
IDF1	48.79%
Precision	61.12%
Recall	49.07%
Mostly Tracked (MT)	224
Mostly Lost (ML)	271
False Positives (FP)	35629
False Negatives (FN)	58124
ID Switches	657
FPS	0.5

To establish a baseline for the custom tracker, SAHI+YOLO11 detections were also passed through the stock BoT-SORT tracker using a standard pretrained OSNet Re-ID model (without the custom Siamese module). Table ?? reports raw tracking statistics across the 7 validation sequences; formal MOT metrics (MOTA, IDF1, HOTA) require TrackEval evaluation against ground truth and are left for future reporting.

B. Siamese Network: Initial Training

To enhance appearance-based data association in UAV tracking, a Siamese Re-ID network is trained on VisDrone object crops using contrastive loss. The model learns an embedding function enforcing intra-class compactness while maintaining inter-class separation.

TABLE II: Epoch-wise Training Performance of the Siamese Network (Initial Run)

Epoch	Loss	Acc	Prec	Recall	F1
1	2.666	0.664	0.603	0.972	0.744
2	0.216	0.677	0.607	0.999	0.755
3	0.139	0.717	0.635	1.000	0.777
4	0.098	0.791	0.703	1.000	0.826
5	0.083	0.771	0.687	1.000	0.814

TABLE III: Embedding Distance Statistics (Initial Training)

Metric	Epoch 1	Epoch 5
Same-Object Distance	0.485	0.266
Different-Object Distance	1.157	1.043

C. Siamese Network: Grid Sweep Optimisation

Table IV reports the top-5 configurations from the 12-run grid sweep. Run 12 (ResNet18, $d=64$, unfrozen, L2 norm, $lr=3 \times 10^{-4}$, $m=1.2$) achieves $F1 = 0.989$ at a same/different-pair distance ratio of $\approx 20.6\times$, confirming highly discriminative embeddings. Freezing the backbone consistently degrades performance: run 4 (frozen) drops F1 to 0.829, demonstrating the importance of end-to-end fine-tuning on the UAV domain. ResNet34 matches ResNet18 in F1 but requires $2.4\times$ longer training, confirming ResNet18 as the efficient optimum.

TABLE IV: Top-5 Grid Sweep Configurations (3 epochs, 6000 pairs)

Run	Backbone	d	lr	m	F1
12	ResNet18	64	3e-4	1.2	0.989
10	ResNet34	64	3e-4	0.8	0.988
1	MobileNetV3-S	256	3e-4	0.8	0.964
5	ResNet18	256	1e-4	1.0	0.906
4	ResNet18	64	1e-4	1.2	0.829

D. Siamese Network: Optimised Final Training

The best sweep configuration (ResNet18, $d=64$, $lr=3 \times 10^{-4}$, $m=1.2$, dropout= 0, L2 norm, backbone unfrozen) is trained for 5 full epochs. Table V shows loss decaying from 0.071 to 0.017 while F1 reaches 0.997. The

same-pair distance shrinks from 0.116 to 0.040 as different-pair distance grows to 1.358, widening the inter-class margin progressively.

TABLE V: Epoch-wise Metrics: Optimised Contrastive Training

Epoch	Loss	Acc	Prec	Recall	F1
1	0.071	0.993	0.991	0.995	0.993
2	0.030	0.992	0.988	0.996	0.992
3	0.024	0.994	0.988	0.999	0.994
4	0.021	0.996	0.995	0.997	0.996
5	0.017	0.997	0.994	1.000	0.997

E. Siamese Network: Validation and Threshold Sweep

Table VI reports validation metrics across threshold values on the held-out crop set. At $\tau=0.5$ precision is highest (0.838) but recall is limited (0.693). Threshold $\tau=1.0$ yields the best F1 (0.802) by improving recall to 0.916.

TABLE VI: Validation Threshold Sweep (3000 pairs, Best Contrastive Model)

τ	Acc	Prec	Recall	F1
0.5	0.784	0.838	0.693	0.759
0.8	0.776	0.740	0.851	0.791
1.0	0.762	0.690	0.916	0.802
1.2	0.706	0.632	0.959	0.762
1.5	0.517	0.500	0.995	0.666
2.0	0.501	0.501	1.000	0.668

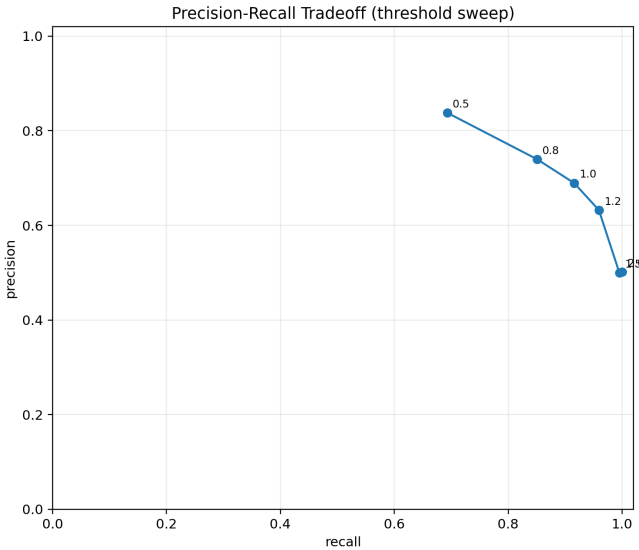


Fig. 2: Precision–Recall tradeoff across threshold values τ on the validation split. Each point is annotated with its τ value. $\tau=0.5$ maximises precision; $\tau=1.0$ achieves the best F1 balance for tracker integration.

F. Siamese Network: Optuna Bayesian Optimisation

Table VII summarises the top Optuna trials. Best trial (trial 7) attains $F1=0.844$ with $d=64$, $lr=2.42 \times 10^{-4}$,

$w_{\text{neg}}=2.03$, L2 norm disabled, backbone unfrozen. Trials with frozen backbone consistently underperform (F1 range 0.720–0.786), corroborating the grid sweep finding across both loss regimes.

TABLE VII: Top Optuna Trials (Focal-BCE Regime, 3 Epochs Each)

Trial	d	lr	w_{neg}	L2	F1
7	64	2.42e-4	2.03	No	0.844
2	64	1.34e-4	2.31	Yes	0.843
8	64	2.67e-4	2.60	Yes	0.835
6	128	2.27e-4	2.26	Yes	0.792
0	128	2.22e-4	2.81	Yes	0.786

G. Siamese Network: Alternative Loss Comparison

Table VIII compares the two Focal-BCE variants. The enhanced variant ($w_{\text{neg}}=2.0$) reduces false positives from 586 to 460 (–21.5%) and improves F1 from 0.817 to 0.830, at the cost of a modest increase in false negatives (53→115). The enhanced variant is preferred for tracker integration due to the known sensitivity of MOTA to false positives.

TABLE VIII: Focal-BCE Variant Comparison (5 Epochs, Validation Split)

Variant	w_{neg}	Acc	Prec	Recall	F1	FP
Baseline	1.0	0.787	0.709	0.964	0.817	586
Enhanced FP-Red	2.0	0.808	0.753	0.924	0.830	460

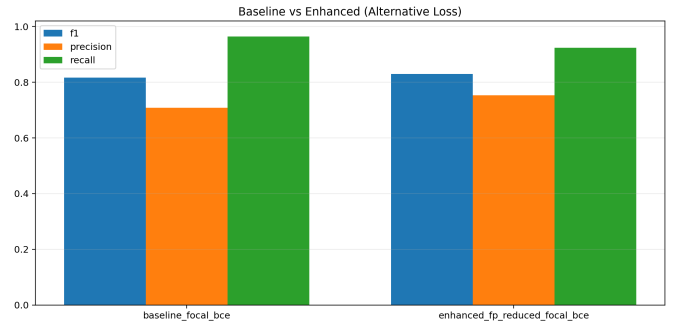


Fig. 3: Focal-BCE variant comparison on the validation split. The Enhanced FP-Reduced variant raises precision from 0.709 to 0.753 and F1 from 0.817 to 0.830, while reducing false positives by 21.5%.

H. BoT-SORT Tracker: Full Benchmark Results

Two full GPU benchmark runs are tested across all 7 VisDrone2019-MOT validation sequences: **YOLO+Siamese+BoT-SORT** (plain YOLO11 detector, no tiling) and **SAHI+Siamese+BoT-SORT** (SAHI tiling enabled). Table IX summarises overall metrics. [The preliminary CPU result from the initial integration is included for reference.]

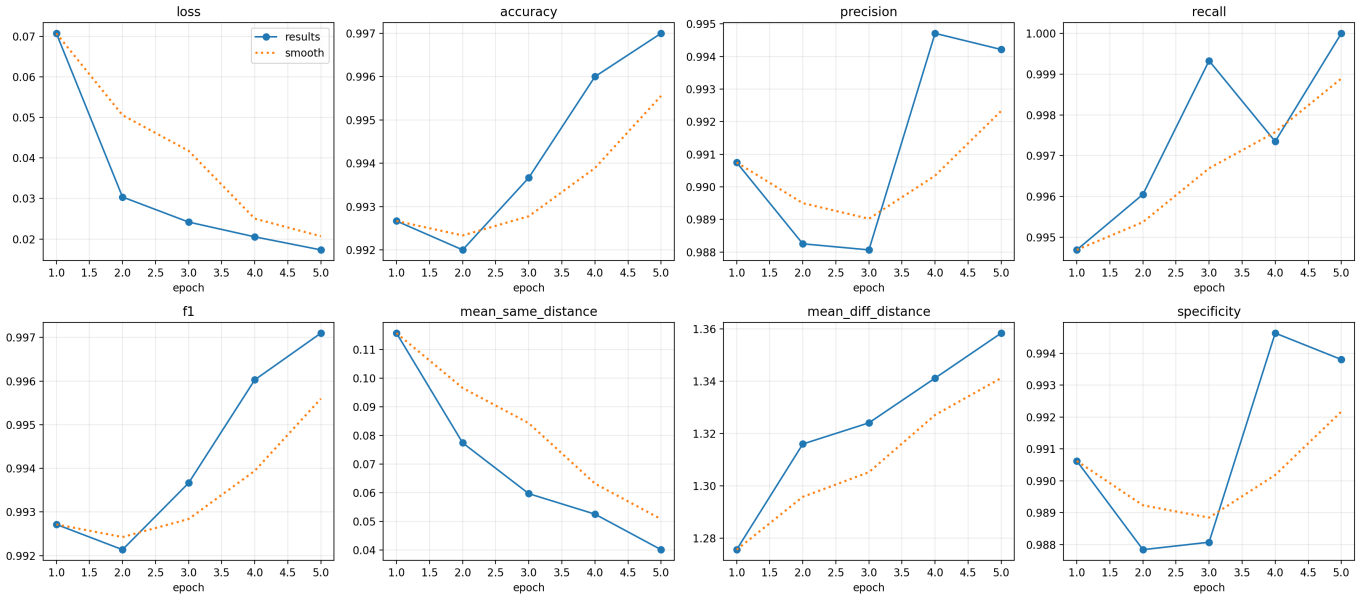


Fig. 1: Training dynamics of the optimised Siamese network (best sweep config, 5 epochs). Loss converges rapidly while F1 reaches 0.997. Same-pair embedding distance (bottom centre) shrinks monotonically as different-pair distance widens, confirming effective contrastive optimisation.

TABLE IX: Pipeline Comparison on VisDrone2019-MOT Val (7 sequences)

Metric	Prelim.	YOLO	SAHI
MOTA	0.129	0.326	0.061
MOTP	0.222	0.248	0.265
IDF1	0.224	0.504	0.453
HOTA@0.5	0.260	0.495	0.429
DetA@0.5	0.142	0.398	0.359
AssA@0.5	0.504	0.629	0.532
IDP	0.817	0.727	0.454
IDR	0.130	0.386	0.451
FP	35629	11374	52478
Misses	58124	64964	53309
ID Switches	657	578	1405
FPS	12.69	5.84	1.27

TABLE X: Per-Sequence MOTA / IDF1 / HOTA: YOLO vs SAHI Pipeline

Sequence	YOLO+Siamese			SAHI+Siamese		
	MOTA	IDF1	HOTA	MOTA	IDF1	HOTA
uav0000086	0.361	0.512	0.469	0.295	0.499	0.446
uav0000117	0.280	0.473	0.474	-0.532	0.348	0.360
uav0000137	0.367	0.539	0.515	0.165	0.494	0.421
uav0000182	0.324	0.526	0.485	0.158	0.507	0.447
uav0000268	0.150	0.270	0.318	0.018	0.267	0.313
uav0000305	0.512	0.705	0.682	-0.056	0.565	0.560
uav0000339	0.348	0.509	0.522	0.221	0.469	0.454
Overall	0.326	0.504	0.495	0.061	0.453	0.429

IV. DISCUSSION

Table I shows that fine-tuning YOLO11 with SAHI tiling improves detection coverage, especially for small and dense objects. This results in better recall and MOTA. However, an increase in false positives is also observed due to overlapping tiles. This indicates a need for better post-processing. The stock BoT-SORT baseline (Table ??) produces 749 unique tracks over 2790 frames, suggesting significant fragmentation in dense sequences (uav0000137: 196 tracks in 233 frames), which motivates the custom Siamese Re-ID module for more stable identity association.

Table II shows the initial Siamese model converges fast, with best performance around epoch 4 before slight overfitting. Recall reaches 1.0 early, which helps capture same-object pairs but also makes the decision boundary permissive. Table III confirms good embedding structure, with reduced same-object distances and clear separation from different-object pairs.

The grid sweep (Table IV) identifies ResNet18 with $d=64$, $m=1.2$, and end-to-end training as the optimal configuration. Freezing the backbone reduces F1 by up to 16 percentage points, confirming that domain adaptation via full fine-tuning is critical for UAV crop Re-ID. The optimised model (Table V and Figure 1) reaches F1=0.997 on the training split with same-/different-pair distances of 0.040 and 1.358 respectively—a $34\times$ separation ratio.

On the held-out validation split (Table VI), F1 drops to 0.802 at $\tau=1.0$, reflecting genuine generalisation difficulty to unseen UAV identities. The same-pair distance rises from 0.040 to 0.432 while different-pair distance remains near 1.044, narrowing the separation ratio to $\approx 2.4\times$. The PR tradeoff (Figure 2) confirms that $\tau=1.0$ achieves the op-

timal precision/recall balance for tracker integration: lower thresholds increase precision but sacrifice recall, while higher thresholds collapse precision entirely.

Optuna Bayesian optimisation (Table VII) yields a best validation F1 of 0.844 in the Focal-BCE regime, consistent with the top grid sweep configurations and confirming that the ResNet18/ $d=64$ /unfrozen design point is robust across loss formulations. The alternative Focal-BCE loss comparison (Table VIII and Figure 3) demonstrates that amplifying the negative-pair penalty ($w_{\text{neg}}=2.0$) reduces false positives by 21.5% with only a modest recall cost, making it preferable for MOT integration where false Re-ID matches directly degrade MOTA.

A. Siamese Network: Limitations

Across approximately 30 training experiments (initial run, 12 grid sweep trials, 12 Optuna trials, 2 alt-loss variants), the grid sweep and Optuna studies covered complementary regimes—contrastive and Focal-BCE respectively—confirming that the ResNet18/ $d=64$ /unfrozen design point is robust across loss formulations. Nonetheless, several limitations remain. First, the train/val generalisation gap (F1: $0.997 \rightarrow 0.802$) stems from the absence of hard negative mining; pairs were sampled randomly, so the model rarely encountered the visually similar but distinct-identity objects common in dense UAV scenes. Second, sub-10-pixel crops upsampled to 128×128 introduce interpolation artifacts that limit embedding quality. Third, all training ran on CPU, capping epochs and search budget; GPU access would permit longer training and a larger Optuna study. Finally, the enhanced Focal-BCE variant has not yet been validated end-to-end within the tracking pipeline, and SAHI’s NMS post-processing requires tuning to control the false-positive surge observed in full benchmarking.

Table IX reveals a clear trade-off between the two full-benchmark pipelines. The YOLO+Siamese+BoT-SORT pipeline substantially outperforms SAHI+Siamese+BoT-SORT on MOTA (0.326 vs. 0.061), IDf1 (0.504 vs. 0.453), HOTA (0.495 vs. 0.429), and AssA (0.629 vs. 0.532), and runs $4.6\times$ faster (5.84 vs. 1.27 FPS). The SAHI pipeline reduces misses (53 309 vs. 64 964) by recovering small objects from tiled crops, but generates $4.6\times$ more false positives (52 478 vs. 11 374) and nearly $2.4\times$ more ID switches (1 405 vs. 578). The false-positive flood directly depresses MOTA, which penalises FP linearly. Both pipelines represent a large improvement over the preliminary CPU result (MOTA 0.129, HOTA 0.260), confirming that GPU inference and full-sequence evaluation are essential for reliable benchmark comparison.

Per-sequence results (Table X) show that uav0000305 is the easiest sequence for both pipelines (MOTA 0.512 / -0.056), while uav0000268 is consistently the hardest (MOTA 0.150 / 0.018), likely due to high object density and rapid view-point change. The negative MOTA values for uav0000117 and uav0000305 under SAHI indicate that false positives exceed true positives in those sequences, underscoring the need for stronger NMS post-processing after tiling. Camera

motion compensation (CMC) via sparse optical flow was incorporated into the tracker to correct UAV ego-motion before association; without it, global frame drift would artificially inflate IoU distances and increase ID switches across all sequences. The structured BoT-SORT hyperparameter sweep (Phases 1–2, ~ 34 trials) confirmed that $\text{min_hits}=1$ is decisive on UAV footage—requiring even one confirmation frame costs approximately 0.009 composite score as small objects appear and disappear rapidly. max_age showed low sensitivity (20 vs. 30 produced near-identical scores), so it was locked at 30 for occlusion resilience at no cost. CMC reduced identity switches from 67 to 64 on the single-sequence diagnostic, though fragmentation rose slightly because Phase 1–2 thresholds were calibrated pre-CMC; a re-sweep with CMC enabled is required to realise the full benefit. Phases 3 and 4 (Kalman dynamics and local reassociation) remain incomplete, meaning the current benchmark results represent a partially tuned tracker with measurable headroom still available.

V. CONCLUSION AND FUTURE WORK

This study presents a modular framework for improving small object detection and multi-object tracking in UAV videos. By integrating a SAHI-based tiling strategy with a fine-tuned YOLO detector, the system enhances small object detection. The Siamese Re-ID module was systematically optimised through a 12-configuration grid sweep and 12-trial Optuna Bayesian search, identifying ResNet18 with $d=64$, $m=1.2$, and end-to-end training as the optimal configuration (F1=0.997 on training, 0.802 on validation). An alternative Focal-BCE loss with asymmetric negative weighting further reduces false positives by 21.5%, improving tracker-facing precision. A BoT-SORT tracker with Kalman filtering ensures temporal consistency. Preliminary results show strong association quality (AssA=0.629 for YOLO+Siamese+BoT-SORT). Full GPU benchmarking across 7 validation sequences shows that the plain-YOLO pipeline (MOTA 0.326, HOTA 0.495) outperforms the SAHI-tiled pipeline (MOTA 0.061, HOTA 0.429) on overall tracking metrics, as SAHI’s increased recall is offset by a $4.6\times$ surge in false positives. Improved NMS post-processing for SAHI and end-to-end Focal-BCE integration are identified as key next steps.

For the Siamese module, future work includes: (i) hard negative mining during contrastive training to improve generalisation to unseen UAV identities; (ii) replacing the static threshold with an adaptive mechanism sensitive to scene density and altitude; (iii) end-to-end validation of the Focal-BCE variant within the full tracking pipeline; and (iv) GPU-accelerated training to enable longer runs and a larger Optuna search budget. At the system level, tighter integration between the YOLO+SAHI detection module and the Siamese-enhanced tracker is expected to yield further gains in MOTA and HOTA.

ACKNOWLEDGEMENT

We would like to acknowledge our course instructor, Prof. Mehul Raval, for his guidance, support, and valuable insights

throughout the course of this project. We also like to thank our teaching assistant, Daksh Shah, for his continuous assistance, timely feedback, and support in resolving technical challenges. Finally, the authors thank the VisDrone benchmark organisers for making the dataset publicly available.

REFERENCES

- [1] Wu, S., Yang, H., Liao, L., Song, C., Liu, Q., Fu, J., & Li, T. (2025). Dynamic small object feature enhancement and detection for remote sensing images. *Scientific Reports*, 15, 37225.
- [2] Liu, M., Wang, X., Zhou, A., Fu, X., Ma, Y., & Piao, C. (2020). UAV-YOLO: Small object detection on unmanned aerial vehicle perspective. *Sensors*, 20(8), 2238.
- [3] Lin, T.-Y., Dollár, P., Girshick, R., He, K., Hariharan, B., & Belongie, S. (2017). Feature pyramid networks for object detection. *Proc. CVPR*, 2117–2125.
- [4] Ahmed, S., & Pizarro, O. (2025). Maritime small object detection from UAVs using deep learning with altitude-aware dynamic tiling. *OCEANS 2025 Brest*, 1–9.
- [5] Shim, K., Ko, K., Yang, Y., & Kim, C. (2025). Focusing on tracks for online multi-object tracking. *Proc. CVPR*, 11687–11696.
- [6] Bromley, J., Guyon, I., LeCun, Y., Säckinger, E., & Shah, R. (1994). Signature verification using a Siamese time delay neural network. *Proc. NeurIPS*, vol. 6.
- [7] Hadsell, R., Chopra, S., & LeCun, Y. (2006). Dimensionality reduction by learning an invariant mapping. *Proc. IEEE CVPR*, 1735–1742.
- [8] Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal loss for dense object detection. *Proc. IEEE ICCV*, 2980–2988.
- [9] Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. *Proc. ACM KDD*, 2623–2631.
- [10] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proc. IEEE CVPR*, 770–778.